

Lab 1 Report: Gitea Installation with Docker & Git LFS Support

1. Environment Setup

OS: Windows 11

Docker Desktop: v4.66.1

Gitea: v1.26.1

Git: Latest

Docker Desktop was installed and verified running with "Engine running" status.

2. Part (a): Install Gitea Using Docker

Step 1: Create Project Directory

```
mkdir "%env:USERPROFILE%\gitea-docker"
```

```
cd "%env:USERPROFILE%\gitea-docker"
```

Step 2: Create docker-compose.yml

```
version: "3"
```

```
networks:
```

```
  gitea:
```

```
    external: false
```

```
services:
```

```
  server:
```

```
    image: docker.gitea.com/gitea:1.26.1
```

```
    container_name: gitea
```

```
    environment:
```

```
      - USER_UID=1000
```

```
      - USER_GID=1000
```

```
    restart: always
```

```
    networks:
```

```
      - gitea
```

```
    volumes:
```

```
      - ./gitea:/data
```

```
      - /etc/timezone:/etc/timezone:ro
```

```
      - /etc/localtime:/etc/localtime:ro
```

```
    ports:
```

```
      - "3000:3000"
```

```
      - "222:22"
```

Step 3: Start Gitea

mkdir gitea

docker compose up -d

Step 4: Initialize via Web UI

Accessed http://localhost:3000

Configured SQLite3 database

Set administrator account: XueZhengyang

Completed installation wizard

Result: Gitea successfully deployed and accessible at http://localhost:3000

3. Part (b): Does Gitea Support LFS? Prove It.

Verification Method 1: Check Configuration File

docker exec gitea cat /data/gitea/conf/app.ini | Select-String -Pattern "lfs"

Output:

[lfs]

ENABLED = true

PATH = /data/git/lfs

Verification Method 2: Web Interface

Navigated to Admin Panel -> Configuration

Confirmed LFS module is enabled by default

Verification Method 3: Functional Test (see Part c)

Successfully tracked .bin files with git lfs track

Uploaded 1.1 GB file via LFS protocol

Conclusion: YES, Gitea fully supports Git LFS. The feature is enabled by default in the Docker image and verified through both configuration inspection and functional testing.

4. Part (c): Create a Repo with a Large File (1GB+)

Step 1: Create Repository via Web UI

Repository name: large-file-test

Initialized with README

Step 2: Local Git LFS Setup

```
-----  
git clone http://localhost:3000/XueZhengyang/large-file-test.git  
cd large-file-test  
git lfs install  
git lfs track "*.bin"
```

Step 3: Generate 1GB Test File

```
-----  
fsutil file createnew largefile.bin 1073741824  
(Get-Item largefile.bin).Length / 1GB  
Output: 1 (1GB confirmed)
```

Step 4: Commit and Push

```
-----  
git add .gitattributes largefile.bin  
git commit -m "Add 1GB test file via Git LFS"  
git push origin main
```

Push Output:

```
Uploading LFS objects: 100% (1/1), 1.1 GB | 80 MB/s, done.  
Enumerating objects: 5, done.  
To http://localhost:3000/XueZhengyang/large-file-test.git  
   efc3136..e7fc9c8  main -> main
```

Step 5: Web Verification

```
-----  
File largefile.bin (1.1 GB) visible in repository  
Stored via Git LFS protocol confirmed
```

5. Part (d): Backup and Restore [Optional]

Backup to External Drive/USB

```
-----  
cd "$env:USERPROFILE\gitea-docker"  
docker compose down  
cd "$env:USERPROFILE"  
Compress-Archive -Path "gitea-docker\gitea" -DestinationPath "gitea-backup.zip"  
Copy-Item "gitea-backup.zip" "E:\gitea-backup.zip"
```

Restore on Another Computer

```
-----  
On new machine with Docker installed:  
mkdir "$env:USERPROFILE\gitea-docker"
```

```
cd "$env:USERPROFILE\gitea-docker"
Copy-Item "E:\gitea-backup.zip" .
Expand-Archive -Path "gitea-backup.zip" -DestinationPath "." -Force
Create docker-compose.yml (same as original)
docker compose up -d
```

Result: All repositories, user data, and LFS files fully restored on new machine.

6. Part (e): Install Gitea Without Docker [Optional]

Alternative installation using native binary with MySQL database:

Install MySQL
brew install mysql

Create database
mysql -u root -e "CREATE DATABASE gitea;"
mysql -u root -e "CREATE USER 'gitea'@'localhost' IDENTIFIED BY 'password';"
mysql -u root -e "GRANT ALL PRIVILEGES ON gitea.* TO 'gitea'@'localhost';"

Download and run Gitea binary
wget https://dl.gitea.com/gitea/1.26.1/gitea-1.26.1-linux-amd64
chmod +x gitea
./gitea web

7. Key Screenshots Summary

Screenshot 1: Docker Desktop "Engine running"
Screenshot 2: docker compose ps showing gitea container
Screenshot 3: Gitea installation wizard
Screenshot 4: Gitea homepage after login
Screenshot 5: LFS configuration ENABLED = true
Screenshot 6: large-file-test repository creation
Screenshot 7: git push with LFS upload success (1.1 GB)
Screenshot 8: Web UI showing 1GB LFS file
Screenshot 9: Backup file on external drive (optional)
Screenshot 10: Restored repository on new machine (optional)

8. Conclusion

This lab successfully demonstrated:

1. Docker deployment of Gitea with persistent storage
2. Git LFS support verified through configuration and functional testing
3. Large file handling (1GB+) using LFS protocol
4. Backup/restore capability for data portability
5. Alternative installation methods understood

Gitea proves to be a lightweight, self-hosted Git service with full LFS support suitable for teams requiring large file version control.